Marine A. Denolle

Associate Professor, Earth and Space Sciences, University of Washington

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Google Scholar | scoped6259 | uwgeophysics6888 | denolle-lab.github.io

Research Impact Summary

Research Citations: 3,024 total (h-index: 24, i10: 48) | 0 last-5y (h_5: 0, $i10_5$: 0) | Google Scholar

Publications: 68 peer-reviewed | 6 Nature/Science-family (T1) | 40 AGU-flagship (T2) | 21 domain journals (T3) | 1 preprints

Research Funding: Lead PI: $3.1M (14 grants) | Co-PI/Co-I: $4.9M (8 grants) | Fellowships: $878K (2)

Open-Source Software: 403 GitHub stars | 255 forks | 21 active repos | NoisePy: 210 stars, 83 forks, 19 contributors | 684+ PyPI downloads/yr

Mentoring: 8 PhD (4 current, 4 graduated) | 11 postdocs | 16+ undergrads | 10 co-supervised grad students | 5 became faculty

Broader Impact: mlgeo-book: 13 GitHub stars

Education

2008 – 2014 **Stanford University** Geophysics
- Supervisor: Dr. Gregory Beroza
- Co-supervisors: Dr. Eric Dunham, Dr. Jesse Lawrence

2007 – 2008 **Ecole Normale Supérieure - IPGP** Geophysics
- Supervisors: Dr. Satish Singh (IPGP), Dr. David Bercovici (Yale)

2006 **Ecole Normale Supérieure** Earth Sciences

2004 – 2005 **Lycée Chateaubriand, Rennes** Physics-Mathematics

Employment

Applied Environmental Intelligence, Seattle, 2025 – present
Co-Founder

University of Washington, Seattle, WA, 2024 – present
Associate Professor
Department of Earth and Space Sciences

University of Washington, Seattle, WA, 2021 – 2024
Assistant Professor
Department of Earth and Space Sciences

Harvard University, Cambridge, MA, 2016 – 2021
Assistant Professor
Department of Earth and Planetary Sciences

UC San Diego, Scripps Institution of Oceanography, La Jolla, CA, 2014 – 2016
Green Postdoctoral Fellow
Institute of Geophysics and Planetary Physics
- Supervisor Dr. Peter Shearer

Awards

2023 **Invited Professorship** Ecole Normale Supérieure rue d'Ulm-Paris

2023 **Data Science Fellow** eScience Institute, University of Washington

2019 **Charles F. Richter Early Career Award** Seismological Society of America

2019 **Kavli Frontiers of Science Fellow** National Academy of Sciences

2019 **Radcliffe Assistant Professorship** Institute for Advanced Study

2018 **CAREER Award** NSF

2017 **David and Lucile Packard Foundation Fellowship** David and Lucile Packard Foundation

2016 **Outstanding Reviewer** Geophysical Research Letters

2015 **Outstanding Reviewer** Geophysical Journal International

2012 **AGU Outstanding Student Paper Award** American Geophysical Union

2012 **SSA Student Presentation Award** Seismological Society of America

2010 **AGU Outstanding Student Paper Award** American Geophysical Union

Teaching

ESS 469/569
U of Washington
2021 - 2024

Machine Learning in the Geosciences
geo-smart.github.io/mlgeo-book

- Taught Fall 2021, 2022, 2023, 2024
- Open-access Jupyter Book with asynchronous teaching materials
- Course adopted by University of Arizona and UC Berkeley
- Curated geoscience-specific ML datasets for homework

ESS 563
U of Washington
2023 - 2025

Computational/Advanced Seismology
denolle-lab.github.io/teaching/ess563

- Taught Spring 2023, 2024, 2025
- Covers ambient noise seismology, earthquake source characterization
- Julia and Python programming for high-performance computing

ESS 412/512
U of Washington
2023 - 2026

Introduction to Seismology
denolle-lab.github.io/teaching/ess412

- Taught Winter 2023, 2025, 2026
- Covers earthquake physics, seismic waves, and hazard assessment

ESS 314
U of Washington
2022 - 2026

Geophysics
uw-geophysics-edu.github.io/ess314

- Taught 2022, 2023, 2024, 2026

EPS 268
Harvard
2019

Machine Learning in Earth and Planetary Sciences

- Taught Fall 2019

EPS 268
Harvard
2018

Induced Seismicity

- Taught Fall 2018

EPS 203
Harvard
2018 - 2020

Earthquakes and Faulting

- Taught Fall 2018, 2020

EPS 55
Harvard
2017 - 2020

Earthquakes and Tectonics

- Taught Fall 2017, 2020

EPS 204
Harvard
2016

Earthquake Sources

- Taught Spring 2016

Phd Advisees

PhD Pre-Candidate
UW ESS
2025 - present

Michael Hemmett

Offshore geophysics

PhD Candidate
USS
2022 - present

Manuela Koepfli

Geohazard assessment and early warning systems

- 1 publication, 1 in review, 1 in prep

PhD Pre-Candidate
UW ESS
2022 - present

Akash Kharita

Geohazard characterization using machine learning

- 2 publications

PhD Candidate
UW ESS
2021 - present

Yiyu Ni

Machine learning and big data seismology

- 8 publications

PhD
Harvard
2019 - 2024

Congcong Yuan

Time-dependent seismology, solid-fluid interaction. Next position: Cornell postdoc ⟶ Faculty position at NTU Singapore

- 4 publications

PhD
Harvard
2018 - 2023

Stephanie Olinger

Cryo-seismology (50% co-advised with Brad Lipovsky). Next position: Stanford Thompson Postdoctoral Fellowship ⟶ CEO of AEI

- 4 publications

PhD
Harvard
2016 - 2021

Tim Clements

Hydro-seismology and big-data seismology. Next position: USGS Mendenhall Postdoc ⟶ USGS Research Geophysicist

- 4 publications

PhD
Harvard
2016 - 2022

Jiuxun Yin

Earthquake seismology and source characterization. Next position: Caltech SCSN Postdoc Fellowship ⟶ Schlumberger

- 6 publications

Postdocs

University of Washington
2022 - 2024
Dr. Qibin Shi
DAS, Deep Learning, Earthquake Catalog building, AgroSeismology
• Now Postdoctoral Fellow at Rice University

University of Washington
2023 - 2024
Dr. Kuan-Fu Feng
Environmental Seismology, Cloud computing
• Next position: Postdoc at University of Utah

University of Washington
2023 - 2025
Dr. Ethan Williams
DAS
• Next position: Assistant Professor at UC Santa Cruz

University of Washington
2024 - 2025
Dr. Stephanie Olinger
Cryoseismology, DAS, Ambient noise monitoring
• Next position: Climate Tech ⟶ CEO Applied Environmental Intelligence

Harvard University/University of Washington
2020 - 2022
Dr. Laura Ermert
Environmental seismology
• Next position: Assistant Professor at ISTerre

Harvard University
2019 - 2020
Dr. Xiaotao Yang
Ambient noise cross correlation
• Next position: Assistant Professor at Purdue

Harvard University
2019 - 2020
Dr. Kurama Okubo
Ambient noise monitoring
• Next position: Researcher at NIED, Japan

Harvard University
2019 - 2020
Dr. Zhitu Ma
Ambient noise cross correlation
• Next position: Assistant Professor at Tongji University, China

Harvard University
2018 - 2019
Dr. Chengxin Jiang
Ambient noise seismology, computational seismology
• Next position: Research Associate at Australian National University

Harvard University
2016 - 2017
Dr. Chris Van Houtte
Earthquake Source seismology

Harvard University
2016 - 2018
Dr. Loïc Viens
ambient noise seismology, ground motion prediction
• Next position Researcher at Los Alamos

Other Graduate Student Supervision

Note: (*) At UW, roles are secondary advisor or primary advisor on one project/manuscript. (**) Advising resulted in a peer-reviewed publication.

Maleen Kidiwela (*,):** 2021-present, UW Oceanography (co-advised with William Wilcock) - DAS and offshore seismology, resulted in publication

Zoe Krauss (*,):** 2021-2024, UW Oceanography (co-advised with William Wilcock) - Axial Seamount hydrothermal monitoring, resulted in publication

Parker Sprinkle (*,):** 2021-present, UW ESS (co-advised) - Earthquake early warning and machine learning

Natasha Toghramadjian (*,):** 2018-2025, Harvard EPS - Seismic attenuation and site effects, resulted in publications

Zhuo Yang (*,):** 2018-2021, Harvard EPS - Machine learning for seismic phase picking, resulted in publication

William Flanagan (*,):** 2019, Harvard EPS - Volcano seismology project

Thibault Pérol (*,):** 2017, Harvard EPS - Deep learning for earthquake detection, resulted in multiple publications

Congcong Yuan (*,) :** 2019, USTC China - Master student visiting researcher on time-dependent seismology, resulted in publication

Philippe Danré (*,):** 2018, Ecole Normale Supérieure Paris - Master thesis on seismic monitoring, resulted in publication

Manuel Florez (*,):** 2017-2019, MIT - Dissertation committee member for geophysics PhD

Graduate Student Committees

Koepfli, Manuela: Earth and Space Sciences, Chair, 2024-present

Ni, Yiyu: Earth and Space Sciences, Chair, 2024-present

Kharita, Akash: Earth and Space Sciences, Advisor, 2023-present

DeGrande, Jensen: Earth and Space Sciences, Member, 2023-present

Pearson, Anna Elaine Rogers: Earth and Space Sciences, Member, 2024-present

Sprinkle, Parker: Earth and Space Sciences, Member, 2022-present

Kidiwela, Maleen: School of Oceanography, Member, 2024-present

Krauss, Zoe: School of Oceanography, GSR, 2022-2024

Sangmin, Song: School of Oceanography, GSR, 2024-present

Zhang, Maochuan: School of Oceanography, GSR, 2023-present

Jones, Randall: Atmospheric Sciences, GSR, 2025-present

Chien, Mu-Ting: Atmospheric Sciences, GSR, 2023-2024

Sweeney, Aodhan: Atmospheric Sciences, GSR, 2024-present

Ragland, John: Electrical and Computer Engineering, GSR, 2023-2024

Rasanen, Ryan: Civil and Environmental Engineering, GSR, 2022-2023

Velappan, Hemalatha: School of Environmental and Forest Sciences, GSR, 2023-present

Velegar, Meghana S: Applied Mathematics, GSR, 2023-2023

Zahn, Olivia: Physics, GSR, 2022-2024

International Phd Advising

Note: Dissertation and defense evaluating committee members

Marius Paul Isken : 2024 (GFZ, Germany)

Luc Illien : 2023 (GFZ, Germany)

Zoe Renate : 2023 (Universite de Lorraine, France)

Reza Esfahani : 2022 (GFZ, Germany)

Kurama Okubo : 2019 (ENS, Paris, France)

Media Coverage

- 2025 Global-scale Database of Seismic Phases
- 2024 Ambient Field Monitoring in the Shallow Crust
- 2024 Making Big Data Look Small
- 2023 UW eScience Seminar
- 2023 Open Science in Seismology
- 2023 UW News Fiber Sensing
- 2018 Radcliffe Seminar: Where we stand in Earthquake Prediction

Undergraduate Students

Note: My undergraduate advising includes research opportunities through summer programs and independent studies (ESS 499 for credit during academic year, hourly pay during summer/breaks). I mentor students toward conference presentations and publications, and support competitive fellowship applications. Legend: (*) Conference presentation | (**) Peer-reviewed publication | (***) Publication in prep | (+) NSF GRFP recipient

Derek Yao: 2026-present, CSE - various agents for Geosciences

Alex Rose : 2025-present, UW Oceanography - DAS data processing research

Anjani Mirchandi: 2024-present, UW Applied Math - DAS data processing research

Hiroto Bito: 2023-2026, UW ESS - ML for earthquake catalog building and spatio-temporal forecasting

Nicholas Wolfe: 2023, UW ESS - Earthquake magnitude estimation project

Informatics Capstone Team (lead Rona Guo(*)): 2023, Nathan Limono, William Phan, Michael Yung, Matthew Herradura), UW Informatics - DAS Platform development, resulted in conference presentation

Lucas Swanson : 2023, UW Informatics - DAS software development

Francesca Skene (*) : 2022-2023, UW ESS - Surface event catalog, resulted in conference presentation

Nick Smoczyk (*,*) :** 2022-2024, University of Minnesota - Volcano seismology using RedPy catalog, resulted in conference presentation and publication in prep

Julian Schmitt (*,+) : 2020-2021, Harvard EPS - Ambient noise seismology on cloud computing with Julia, BASIN project, resulted in conference presentation and NSF GRFP

Jared Bryan (*,,+):** 2019, Harvard EPS - Ambient noise monitoring, SCEC Internship, resulted in conference presentation, publication, and NSF GRFP

Albert Aguilar (*): 2018, Harvard EPS - Subduction zone seismology and data mining, SCEC, resulted in conference presentation

Leore Lavin: 2016, Harvard EPS - Ground motion simulations with ambient noise

Roy Bowling (*): 2014, Scripps IGPP - Ambient noise seismology, SCEC Internship

Tara Larrue (*): 2012, Stanford Geophysics - Ambient noise seismology, SURGE program

Penprapa Wutthijuk (*): 2011, Stanford - Ambient noise seismology, SURGE program

Training Workshops

2025 **CRESCENT-SCOPED Workshop** co-PI organizer for ML and Earthquake Catalog Building workshop (35 participants)

2024 **SCOPED Workshop** Lead PI organizer for Cloud/HPC/wavefield simulations workshop (100 participants)

2024 **SSA Cloud 101 & Data Mining** Lead organizer of cloud workshop (80 participants)

2023 **CyberTraining Workshop** Lead PI, coordinator, and instructor for HPC and Data Science

2021 – present **GeoSMART ML for Solid Earth Science** Co-organizer of ML training with Jupyter Book

2016 **SCEC-ERI VISES Summer School** Scientific committee member and instructor

University Service

University of Washington
2026 – present
AI Minor task force member
- reviewing AI Minor proposal for cross-university program, ~ 1hr/week for spring quarter

University of Washington
2026 – present
Department Colloquium Committee
- Help organizing the department colloquium series

University of Washington
2026 – present
AI@UW Advisory Committee to the VP
- Advising AI strategy and initiatives across the university, commitment ~ 1hr/week

University of Washington
2024 – present
Co-director of UW CS4Env
- Computer Science for the Environment (CS4Env) is a cross-unit initiative to promote synergies between computer science and environmental research at UW, includes organizing biweekly seminar, reviewing proposals, and organizing an annual symposium.

UW/ESS
2024 – present
Chair of Curriculum Committee
- Reviewing proposed courses, course changes, undergraduate and graduate program.

UW/ESS
2022 – present
Member of Curriculum Committee and Data Science Oversight Committee
- Review the proposed data science option program

UW/ESS
2022 – 2023
Member of Executive Committee

UW/ESS
2022 – 2023
Member of Research Faculty Search Committees
- Search committee for PNSN Network Manager and Igneous Processes Faculty Search

Harvard University
2016 – 2020
Multiple Committee Memberships
- Undergraduate Curriculum Committee, Graduate Student Council, IT Committee, Diversity Inclusion and Belonging Committee, Department Colloquium Committee

Professional Service

2026 – present **Member of Board of Directors at Earthscope Consortium**

2023 – 2025 **Chair of Integration and Innovation Advisory Committee at Earthscope Consortium**

2024 – 2025 **Member of Science Steering Committee at SCEC**

2021 – 2022 **Member of Data Service Standing Committee at IRIS**

2021 – 2023 **Member of HPC Standing Committee at SCEC**

2011 – 2012 **Chair of Graduate Student Council at Stanford University**

2022 – 2022 **Committee Member of Charles Richter Early Career Award Committee at Seismological Society of America**

Review Panels

2020,2024 **NSF Review Panels EAR-Geophysics and GeoInformatics**

- Reviewed 10+ proposals per panel

2016 – 2018 **USGS Review Panels Earthquake Hazards Program**

- Reviewed 10+ proposals per panel

Editorial Service

2025 – present **Editor at Geophysical Journal International**

2017 – 2020 **Associate Editor at Geophysical Research Letters**

2014 – present **Reviewer for multiple journals in geophysics and seismology**

- >150 reviews for GJI, BSSA, Nature Communications, GRL, JGR, Science, and others

Publications Legend

Author Notation: Bold: Marine A. Denolle (PI). *Italic*: graduate students, postdocs, and undergraduate students mentored by MD.

Publications

2026 Veronica Gaete-Elgueta, *Manuela Köpfli*, Dominik Gräff, Bradley P Lipovsky, **Marine A. Denolle**, Weston A. Thelen, Brendan Pratt, Taylor R. Kenyon (2026). **Distributed Acoustic Sensing Records of Earthquakes and Surface Processes at Mount Rainier Volcano**. in review in Seismica

2026 *Qibin Shi*, David R. Montgomery, Abigail L.S. Swann, Nicoleta C. Cristea, *Ethan F. Williams*, Nan You, Simon Jeffery, Joe Collins, Ana Prada Barrio, Paula A. Misiewicz, Tarje Nissen-Meyer, **Marine A. Denolle** (2026). **Agroseismology and the impact of farming practices on soil hydrodynamics**. Science, 392, 10.1126/science.aec0970
Featured in: UW News, Grist, Dailyuw

2026 *Maleen Kidiwela*, **Marine A. Denolle**, William S. D. Wilcock, *Kuan-Fu Feng* (2026). **Active protothrusts and fluid highways: Seismic noise reveals hidden subduction dynamics in Cascadia**. Science Advances, 12, 10.1126/sciadv.aea3684
Featured in: Npr, UW News

2026 Han Xiao, Frederik Tilmann, Martijn van den Ende, Diane Rivet, Afonso Loureiro, Takeshi Tsuji, Arantza Ugalde, *Qibin Shi*, **Marine A Denolle** (2026). **DeepSubDAS: an earthquake phase picker from submarine distributed acoustic sensing data**. Geophysical Journal International, 245, 10.1093/gji/ggag061

2026 *Akash Kharita*, **Marine Denolle**, Alexander Hutko, Renate Hartog, Stephen Malone (2026). **Exploration of Machine Learning Methods to Seismic Event Discrimination in the Pacific Northwest**. Seismica, 5, 10.26443/seismica.v5i1.2068

2026 John Townend, Ilma del Carmen Juarez-Garfias, Olivia Pita-Sllim, Calum J. Chamberlain, Emily Warren-Smith, Caroline Holden, Kasper van Wijk, **Marine Denolle**, Andrew Curtis, Hiroe Miyake (2026). **Seismological Analysis of Contemporary and Future Alpine Fault Earthquakes Using the Southern Alps Long Skinny Array (SALSA)**. Seismological Research Letters, 10.1785/0220250260

2026 *Natasha Toghramadjian*, **Marine A. Denolle**, *Laura Ermert*, *Chengxin Jiang* (2026). **Probing the Seattle Basin Edge Using a Dense Urban Nodal Array in 100 Backyards**. Seismological Research Letters, 10.1785/0220250241

2026 Alexey Yermakov, Yue Zhao, **Marine Denolle**, *Yiyu Ni*, Philippe M. Wyder, Judah Goldfeder, Stefano Riva, *Jan Williams*, David Zoro, Amy Sara Rude, Matteo Tomasetto, Joe Germany, Joseph Bakarji, Georg Maierhofer, Miles Cranmer, J. Nathan Kutz (2026). **The Seismic Wavefield Common Task Framework**. arXiv, 10.48550/arXiv.2512.19927

2026 *Kuan-Fu Feng*, **Marine Denolle**, Fan-Chi Lin, Tonie van Dam (2026). **A Decadal Survey of the Near-Surface Seismic Velocity Response to Hydrological Variations in Utah, United States**. Journal of Geophysical Research: Solid Earth, 131

2025 *Yiyu Ni*, **Marine Denolle**, Amanda Thomas, Alex Hamilton, Jannes Münchmeyer, Yinzhi Wang, Loïc Bachelot, Chad Trabant, David Mencin (2025). **A Global-scale Database of Seismic Phases from Cloud-based Picking at Petabyte Scale**. Seismica, 4, 10.26443/seismica.v4i2.1738
Featured in: Video

2025 *Yiyu Ni*, **Marine A Denolle**, Jannes Münchmeyer, Yinzhi Wang, *Kuan-Fu Feng*, Carlos Garcia Jurado Suarez, Amanda M Thomas, Chad Trabant, Alex Hamilton, David Mencin (2025). **A Review of Cloud Computing and Storage in Seismology**. Geophysical Journal International, 10.1093/gji/ggaf322

2025 **Marine A. Denolle**, Carl Tape, Ebru Bozdağ, Yinzhi Wang, Felix Waldhauser, Alice-Agnes Gabriel, Jochen Braunmiller, Bryant Chow, Liang Ding, *Kuan-Fu Feng*, Ayon Ghosh, Nathan Groebner, Aakash Gupta, *Zoe Krauss*, Amanda M. McPherson, Masaru Nagaso, Zihua Niu, *Yiyu Ni*, Rıdvan Örsvuran, Gary Pavlis, Felix Rodriguez-Cardozo, Theresa Sawi, David Schaff, Nico Schliwa, David Schneller, *Qibin Shi*, Julien Thurin, Chenxiao Wang, Kaiwen Wang, Jeremy Wing Ching Wong, Sebastian Wolf, *Congcong Yuan* (2025). **Training the Next Generation of Seismologists: Delivering Research-Grade Software Education for Cloud and HPC Computing Through Diverse Training Modalities**. Seismological Research Letters, 96, 10.1785/0220240413

2025 **Marine A. Denolle**, *Qibin Shi*, *Tim Clements*, *Loïc Viens*, Veronica Rodriguez-Tribaldos, Fabrice Cotton (2025). **Ambient field seismology in critical zone hydrological sciences**. Comptes Rendus. Géoscience, 357, 10.5802/crgeos.310

2025 *Q. Shi*, **M. A. Denolle**, *Y. Ni*, *E. F. Williams*, N. You (2025). **Denoising Offshore Distributed Acoustic Sensing Using Masked Auto-Encoders to Enhance Earthquake Detection**. JGR: Solid Earth, 130, 10.1029/2024JB029728

2025 *Q. Shi*, *E. F. Williams*, B. P. Lipovsky, **M. A. Denolle**, W. S. D. Wilcock, D. S. Kelley, K. Schoedl (2025). **Multiplexed Distributed Acoustic Sensing Offshore Central Oregon**. Seismological Research Letters, 96, 10.1785/0220240460
Featured in: UW News, Phys

2024 *Y. Ni*, **M. A. Denolle**, *Q. Shi*, B. P. Lipovsky, S. Pan, J. N. Kutz (2024). **Wavefield reconstruction of distributed acoustic sensing: Lossy compression, wavefield separation, and edge computing**. Journal of Geophysical Research: Machine Learning and Computation, 1, 10.1029/2024JH000247

2024 P. Makus, **M. A. Denolle**, C. Sens-Schönfelder, *M. Köpfli*, F. Tilmann (2024). **Analysing Volcanic, Tectonic, and Environmental Influences on the Seismic Velocity from 25 Years of Data at Mount St. Helens**. Seismological Research Letters, 95, 10.1785/0220240088

2024 M. Köpli, **M. A. Denolle**, W. Thelen, P. Makus, S. Malone (2024). **Examining 22 Years of Ambient Seismic Wavefield at Mount St. Helens**. Seismological Research Letters, 95, 10.1785/0220240079

2024 F. Diewald, **M. Denolle**, J. J. Timothy, C. Gehlen (2024). **Impact of Temperature and Relative Humidity Variations on Coda Waves in Concrete**. Scientific Reports, 14, 10.1038/s41598-024-69564-4

2024 *K. Okubo*, B. Delbridge, **M. Denolle** (2024). **Monitoring velocity change over 20 years at Parkfield**. Journal of Geophysical Research: Solid Earth, 129, 10.1029/2023JB028084

2024 T. Cochard, I. Svetlizky, G. Albertini, R. C. Viesca, S. M. Rubinstein, F. Spaepen, *C. Yuan*, **M. Denolle**, Y.-Q. Song, L. Xiao, D. A. Weitz (2024). **Extended crack propagation by local nucleation and rapid transverse expansion**. Nature Physics, 10.1038/s41567-023-02365-0

2023 *A. Kharita*, **M. Denolle**, M. West (2023). **Discrimination between icequakes and earthquakes in southern Alaska: an exploration of waveform features using random forest algorithm**. Geophysical Journal International, 10.1093/gji/ggae106

2023 *S. Olinger*, B. Lipovsky, **M. Denolle** (2023). **Ocean Coupling Limits Rupture Velocity of Fastest Observed Ice Shelf Rift Propagation Event**. AGU Advances, 5, 10.1029/2023AV001023
Featured in: UW News, EOS

2023 *C. Yuan*, T. Cochard, **M. Denolle**, J. Gomberg, A. Wech, L. Xiao, D. Weitz (2023). **Laboratory hydrofracture as analogs to tectonic tremors**. AGU Advances, 5, 10.1029/2023AV001002
Featured in: UW News, EOS

2023 *Q. Shi*, **M. Denolle** (2023). **Improved observations of deep earthquake ruptures using machine learning**. Journal of Geophysical Research: Solid Earth, 128, 10.1029/2023JB027334

2023 *Congcong Yuan*, *Yiyu Ni*, Youzuo Lin, **Marine Denolle** (2023). **Better Together: Ensemble Learning for Earthquake Detection and Phase Picking**. IEEE Transactions on Geoscience and Remote Sensing, 61, 10.1109/TGRS.2023.3320148

2023 *Yiyu Ni*, **Marine A. Denolle**, Rob Fatland, Naomi Alterman, Bradley P. Lipovsky, Friedrich Knuth (2023). **An Object Storage for Distributed Acoustic Sensing**. Seismological Research Letters, 95, 10.1785/0220230172

2023 *Z. Krauss*, *Y. Ni*, S. Henderson, **M. Denolle** (2023). **Seismology in the cloud: guidance for the individual researcher**. Seismica, 2, 10.26443/seismica.v2i2.979

2023 *Y. Ni*, A. Hutko, *F. Skene*, **M. Denolle**, S. Malone, P. Bodin, R. Hartog, A. Wright (2023). **Curated Pacific Northwest AI-ready Seismic Dataset**. Seismica, 2, 10.26443/seismica.v2i1.368

2023 *L. Ermert*, E. Cabral-Cano, E. Chaussard, D. Solano-Rojas, L. Quintanar, D. Morales Padilla, E. A. Fernandez-Torres, **M. A. Denolle** (2023). **Probing environmental and tectonic changes underneath Ciudad de México with the urban seismic field**. Solid Earth (EGU), 10.5194/egusphere-2022-1361

2023 *T. Clements*, **M. A. Denolle** (2023). **The Seismic Signature of California's Earthquakes, Droughts, and Floods**. Journal of Geophysical Research: Solid Earth, 128, 10.1029/2022JB025553

2022 *J. Yin*, **M. A. Denolle**, B. He (2022). **A multitask encoder–decoder to separate earthquake and ambient noise signal in seismograms**. Geophysical Journal International, 231, 10.1093/gji/ggac290

2022 *C. Jiang*, **M. A. Denolle** (2022). **Pronounced Seismic Anisotropy in Kanto Sedimentary Basin: A Case Study of Using Dense Arrays, Ambient Noise Seismology, and Multi-Modal Surface-Wave Imaging**. Journal of Geophysical Research: Solid Earth, 127, 10.1029/2022JB024613

2022 *L. Viens*, *C. Jiang*, **M. A. Denolle** (2022). **Imaging the Kanto Basin seismic basement with earthquake and noise autocorrelation functions**. Geophysical Journal International, 230, 10.1093/gji/ggac101

2022 *S. D. Olinger*, B. P. Lipovsky, **M. A. Denolle**, B. W. Crowell (2022). **Tracking the Cracking: a Holistic Analysis of Rapid Ice Shelf Fracture Using Seismology, Geodesy, and Satellite Imagery on the Pine Island Glacier Ice Shelf, West Antarctica**. Geophysical Research Letters, 10.1029/2021GL097604

2022 *Z. Yang*, *C. Yuan*, **M. A. Denolle** (2022). **Detecting Elevated Pore Pressure due to Wastewater Injection Using Ambient Noise Monitoring**. The Seismic Record, 2, 10.1785/0320210036

2021 *J. Yin*, **M. A. Denolle** (2021). **The Earth's Surface Controls the Depth-Dependent Seismic Radiation of Megathrust Earthquakes**. AGU Advances, 2, 10.1029/2021AV000413
Featured in: EOS

2021 *C. Yuan*, *J. Bryan*, **M. A. Denolle** (2021). **Comparing approaches to measuring seismic phase variations in the time, frequency, and wavelet domains**. Geophysical Journal International, 226, 10.1093/gji/ggab140

2021 *J. Yin*, Z. Li, **M. A. Denolle** (2021). **Source time function clustering reveals patterns in earthquake dynamics**. Seismological Research Letters, 92, 10.1785/0220200403

2021 *T. Clements*, **M. A. Denolle** (2021). **SeisNoise.jl: Ambient Seismic Noise Cross Correlation on the CPU and GPU in Julia**. Seismological Research Letters, 92, 10.1785/0220200192

2020 **M. A. Denolle**, T. Nissen-Meyer (2020). **Quiet Anthropocene, quiet Earth**. Science, 369, 10.1126/science.abd8358

2020 J. P. Jones, *K. Okubo*, *T. Clements*, **M. A. Denolle** (2020). **SeisIO: A Fast, Efficient Geophysical Data Architecture for the Julia Language**. Seismological Research Letters, 91, 10.1785/0220190295

2020 *C. Jiang*, **M. A. Denolle** (2020). **NoisePy: A new high-performance python tool for ambient-noise seismology**. Seismological Research Letters, 91, 10.1785/0220190364

2019 *P. Danré*, *J. Yin*, B. Lipovsky, **M. Denolle** (2019). **Earthquakes Within Earthquakes: Patterns in Rupture Complexity**. Geophysical Research Letters, 43, 10.1029/2019GL083093
Featured in: Harvard Gazette

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2018 *T. Clements*, **M. A. Denolle** (2018). **Tracking groundwater levels using the ambient seismic field**. Geophysical Research Letters, 45, 10.1029/2018GL077706

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2017 Y. Sheng, **M. A. Denolle**, G. C. Beroza (2017). **Multicomponent C3 Green's Functions for Improved Long-Period Ground-Motion Prediction**. Bulletin of the Seismological Society of America, 107, 10.1785/0120170053

2017 *L. Viens*, **M. Denolle**, H. Miyake, S. Sakai, S. Nakagawa (2017). **Retrieving impulse response function amplitudes from the ambient seismic field**. Geophysical Journal International, 210, 10.1093/gji/ggx155

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2016 **M. Denolle**, P. M. Shearer (2016). **New perspective on self-similarity of shallow thrust earthquakes**. Journal of Geophysical Research: Solid Earth, 121, 10.1002/2016JB013105

2015 **M. Denolle**, W. Fan, P. M. Shearer (2015). **Dynamics of the M7.8 2015 Nepal Earthquake**. Geophysical Research Letters, 42, 10.1002/2015GL065336

2014 E.-J. Lee, P. Chen, T. H. Jordan, P. B. Maechling, **M. Denolle**, G. C. Beroza (2014). **Full 3D Tomography (F3DT) for Crustal Structure in Southern California Based on the Scattering-Integral (SI) and the Adjoint-Wavefield (AW) Methods**. Journal of Geophysical Research, 119, 10.1002/2014JB011236

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2014 **M. Denolle**, E. M. Dunham, G. A. Prieto, G. C. Beroza (2014). **Strong Ground Motion Prediction using Virtual Earthquakes**. Science, 343, 10.1126/science.1245678
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2013 **M. Denolle**, E. M. Dunham, G. A. Prieto, G. C. Beroza (2013). **Ground Motion Prediction of Realistic Earthquake Sources Using the Ambient Seismic Field**. Journal of Geophysical Research, 118, 10.1029/2012JB009603

2013 J. F. Lawrence, **M. Denolle**, K. J. Seats, G. Prieto (2013). **A numeric evaluation of attenuation from ambient noise correlation functions**. Journal of Geophysical Research, 118, 10.1002/2012JB009513

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2011 G. A. Prieto, **M. Denolle**, J. F. Lawrence, G. C. Beroza (2011). **On amplitude carried by the ambient seismic field**. Comptes Rendus Geoscience (Thematic Issue: Imaging and Monitoring with Seismic Noise), 343

2010 S. Singh, N. Hananto, A. Chauhan, H. Permana, **M. Denolle**, A. Hendriyana, D. Natawidjaja (2010). **Evidence of active backthrusting at the NE Margin of Mentawai Islands, SW, Sumatra**. Geophysical Journal International, 180, 10.1111/j.1365-246X.2009.04458.x

Grants

2025 **Jerry and Linda Paros** Mult-Sensor, Multi-Geohazards Monitoring at Mt Rainier and Mountaineous regions: $750,000 (lead PI

2025 **UW CRESST - FFST** Toward Predictive Understanding of Climate-Compounded Geohazards: $300,000 (lead PI

2025 **NSF CS4All** Separating the Signal from the Noise: Promoting Alaskan students inquiry with geographically relevant seismic data and machine learning techniques: $889,766 (lead PI Lore, Denolle co-PI, $62,000 to UW), DRL-2524060

2025 **NSF RISE CAIG** Collaborative Research: CAIG: Framework for Artificial Intelligence-Enhanced Modeling of Wildfire Geohazards (FAIM-WG): Applications for postfire debris flows across the Western US: $735,074 (lead PI Istanbulluoglu, Denolle co-PI), RISE-2530591

2025 **NSF OCE** Multi-span distributed fiber sensing on the Ocean Observatories Initiative Regional Cabled Array: $880,490 (lead PI Wilcock, Denolle co-PI), OCE-2526198, technology development grant.

2025 **NSF OCE** EXTension on the ENDeavour Segment (EXTEND): Illuminating the seafloor spreading cycle: $785,819.00 (lead PI Wilcock, Denolle co-PI), OCE-2439442

2024 **NSF EAR** Collaborative Research: A seismic investigation of slow slip and fault locking along the Alaska-Aleutian subduction zone: $248,835.00 (Denolle co-PI, collaborative with lead PI Golos), EAR-2346079

2023 **NSF OCE** RAPID: Multiplexed Distributed Acoustic Sensing (DAS) at the Ocean Observatory Initiative (OOI) Regional Cabled Array (RCA): $198,069.00 (lead PI Wilcock, Denolle co-PI), OCE-2415521

2023 **Ecole Normal Supérieure de Paris** Travel Fellowship: $3,500 (Denolle PI)

2023 **SCEC** CyberTraining for Seismology: Data Science and HPC: $35,229 (Denolle PI)

2021 **David and Lucile Packard Foundation** URG Program: $50,000 (PI)

2021 **Murdock Charitable Trust Fund** UW FiberLab equipment: $950,000 (co-PI, lead PI Brad Lipovsky). Built facility for Fiber Sensing.

2021 **NSF CyberTraining** GeoSMART ML workforce development: $995,817 (co-PI). Build a course MLGEO.

2021 **NSF OAC-CSSI** Collaborative Research: Frameworks: Seismic COmputational Platform for Empowering Discovery (SCOPED): $717,441.00 (Denolle PI, lead-PI Carl Tape), OAC-2103701

2021 **NSF EAR, PREEVENTS** Collaborative Research: Cross-Validation of Empirical and Physics-based ground motion predictions: $167,804.00 (Denolle PI), EAR-2125337

2020 **SCEC grant** Aftershock patterns and co-seismic off-fault damage elucidate dynamic rupture processes on the 2019 Ridgecrest earthquake sequence: $33,307 (Denolle PI) - returned

2019 **Harvard University - David Rockefeller Center for Latin American Studies** Monitoring Seismic Hazards in Mexico City using Grillo, a Low-Cost Earthquake Early Warning System, Hardware, $85,000 (Denolle PI)

2019 **Harvard Data Science Initiative** Ambient-noise seismology using Cloud Computing: $27,210 (Denolle PI)

2018 **NSF EAR, CAREER** CAREER: Dynamics of surface rupturing thrust earthquakes: $504,315 (PI)

2018 **SCEC grant** Data Collection for Virtual Earthquakes on Cajon Pass: $28,085 (Denolle PI) - field work

2017 **David and Lucile Packard Foundation** Fellowship: $875,000 (PI)

2017 **NSF PREEVENTS** Collaborative Proposal - PREEVENTS Track 2: Cascadia Scenario Earthquakes: Source, Path, and implications for Earthquake Early Warning: $324,495.00 (Denolle PI, collaborative with lead PI Yihe Huang and Amanda Thomas), EAR-1739712

2017 **SCEC grant** Static and dynamic source parameters of global strike-slip earthquakes: $25,000 (Denolle PI)

2016 **SCEC grant** Epistemic uncertainties in ground motion prediction from virtual earthquakes: $26,173 (Denolle PI)

2015 **SCEC grant** Basin Response to Virtual Earthquakes on the San Jacinto Fault and the Itoigawa-Shizuoka Fault: $20,000 (Denolle PI)

Research Products

2021 – present **Machine Learning in the Geosciences (Open-Access Jupyter Book)** Textbook
Book and course repository provide graduate-level ML curriculum, asynchronous modules, and homework data pipelines for geoscience training.

- Addresses a core gap: no canonical ML textbook existed for geoscience graduate classrooms.
- Adoption and contribution interest from peer programs (including Arizona and Berkeley).
- Ongoing community development with curated datasets and reproducible teaching workflows.

2019 – present **NoisePy** Software
Open-source Python software for large-scale ambient-noise seismology. GitHub.

- Community scale: 207 stars, 83 forks, 19 contributors (Jan 2026).
- Operational usage: taught in workshops and used for large-scale cross-correlation workflows.
- Open, version-controlled development supporting reproducible and cloud-oriented processing.

2020 – present **SeisNoise.jl** Software
Open-source Julia software for high-performance ambient-noise processing and cross-correlation. GitHub.

- Extends the Julia seismology ecosystem with core ambient-noise tooling.
- Community signal: 50 stars and 17 forks (Apr 2023 snapshot).
- Used by group members for production research workflows.

2023 **EarthML4PNW / PNW AI-Ready Seismic Dataset** Dataset
Curated ML-ready Pacific Northwest dataset product. Dataset repository.

- Publication link: Seismica (2023), DOI 10.26443/seismica.v2i1.368.
- Version-controlled release and metadata stewardship through GitHub workflows.
- Designed for reproducible benchmarking and graduate-level ML coursework.

2025 **Global Seismic Phase Database** Dataset
Global seismic data product from cloud-scale waveform processing. DOI.

- Scale: 4.3 billion phase picks from 1.3 PB of continuous seismic data.
- Publication link: Seismica (2025) with DOI-backed archival record.
- Supports global monitoring and data-driven geophysics workflows.

Invited Talks

2026 **National Academy of Sciences - COSEG** Seminar Speaker

2026 **NGF Science Meeting** Conference Speaker

2026 **WCCM 2026, Munich** Keynote Speaker

2026 **European Geophysical Union 3x** Conference Speaker

2026 **CERI Memphis** Department Colloquium

2025 **IEEE New Era AI World Leader Summit, Bothell** Keynote Speaker

2025 **SIAM Geoscience Meeting** Plenary Speaker

2025 **Workshop on Earthquake Physics and Applications of AI to Tectonic Faulting, Italy** Plenary Speaker

2025 **Civil Environmental Engineering, University of Washington** Department Colloquium

2025 **Radcliffe Institute of Advanced Studies, on the road** Short Talk

2025 **Washington University, Saint Louis** Department Colloquium

2024 **University of Southern California** Department Colloquium

2024 **University of California, Davis** Department Colloquium

2024 **Northern Arizona University** Department Colloquium

2024 **Passive imaging and monitoring in wave physics (Cargese, France)** Invited Conference Talk

2024 **Statewide California Earthquake Center** Plenary Speaker

2023 **Ecole Normale Supérieure, Paris** Séminaire Departemental

2023 **eScience Institute, University of Washington** Data Science Seminar

2023 **Sandia National Lab, GNEM seminar series** Department Colloquium

2022 **American Geophysical Union (x2)** Invited Conference Talk

2022 **University of New Mexico** Department Colloquium

2021 **University of Oregon** Seismo Colloquium

2021 **U Utah, Seismo Tea** Seismo Colloquium

2021 **University of Wisconsin** Department Colloquium

2021 **Colorado School of Mines** Department Colloquium

2021 **Mexico a traves de los sismos** Invited Conference Talk

2020 **University of Washington** Department Colloquium

2020 **UC Berkeley** Department Colloquium

2019 **Yale University** Department Colloquium

2019 **University of Washington, seismolunch** Seismo Colloquium

2019 **EGU annual meeting, Vienna** Invited Conference Talk

2019 **Michigan State University** Department Colloquium

2019 **Victoria University of Wellington, SN Jepson Lecture, New Zealand** Public Lecture

2019 **GNS-Science, New Zealand** Department Colloquium

2019 **Stanford University, Department of Geophysics** Department Colloquium

2019 **Tufts University, Civil Engineering seminar** Department Colloquium

2018 **Brown University** Department Colloquium

2018 **Ecole Normale Superieure, Paris** Department Colloquium

2017 **AGU, New Orleans** Invited Conference Talk

2017 **Columbia University, Lamont Doherty Earth Observatory** Department Colloquium

2016 **Harvard Museum of Natural History** Public Lecture

2016 **University of Oregon** Department Colloquium

2016 **University of New Hampshire, Chapman Colloquium** Department Colloquium

2016 **UC Santa Cruz, IGPP seminar** Department Colloquium

2016 **Massachusetts Institute of Technology** Department Colloquium

2015 **USGS, Menlo Park, Earthquake Hazard Program** Department Colloquium

2015 **University of Victoria, BC, Canada** Department Colloquium

2015 **Penn State, Geodynamics seminar** Department Colloquium

2015 **Harvard, Earth and Planetary Sciences** Department Colloquium

2015 **UT Austin, Solid Earth seminar** Department Colloquium

2015 **UCLA, seismology/tectonics seminar** Department Colloquium

2015 **HOKUDAN International Symposium on Active Faulting, Awaji, Japan** Invited Conference Talk

2015 **Information Theory and Applications workshop, La Jolla** Invited Conference Talk

2015 **IGPP-Scripps Institution of Oceanography, UCSD, Geophysics seminar** Department Colloquium

2015 **University of Southern California, Geophysics seminar** Department Colloquium

2014 **Strong Motion Studies for Future Mega-Quakes, DPRI, Kyoto University, Japan** Invited Conference Talk

2014 **AGU-SEG Summer Research workshop, Vancouver, Canada** Invited Conference Talk

2014 **San Diego State University** Department Colloquium

2014 **UC Santa Barbara** Department Colloquium

2014 **IGPP-Scripps Institution of Oceanography, UCSD, Geophysics seminar** Department Colloquium

2013 **AGU, Meeting of the Americas, Cancun, Mexico** Invited Conference Talk

2013 **Caltech Seismo Lab** Department Colloquium

2013 **USGS, Menlo Park** Department Colloquium

2013 **Stanford ICME seminar** Department Colloquium

2013 **Earthquake Research Institute, Tokyo University, Japan** Department Colloquium

2013 **Advanced Industrial Science and Technology, Japan** Department Colloquium

2013 **Disaster Prevention Research Institute, Japan** Department Colloquium

2012 **ACOUSTICS, France** Invited Conference Talk

2012 **Berkeley Seismo Lab** Department Colloquium

2011 **Institut de Physique du Globe de Paris, Earthquake seminar** Department Colloquium